

UNIT TWO

THE HEALTH BENEFITS OF PHYSICAL ACTIVITY

Unit objectives

By the end of this unit students should be able to:

- Describe the health benefits of physical activity
- Identify diseases which are associated with a sedentary lifestyle and the major risk factors for these diseases
- Realize regular physical activity in terms of disease prevention and healthy aging?
- Estimate physical activity needed for improved health benefits?
- Identify physical activities suitable for typical people, and how often should they exercise?

INTRODUCTION

- Physical activity and physical fitness have been linked with health and longevity since ancient times.
- The earliest records of organized exercise used for health promotion are found in China, around 2500 BC.
- the Greek physicians of the fifth and early fourth centuries BC who established a tradition of maintaining positive health through 'regimen'; the combination of correct eating and exercise

Hippocrates (460–370 BC), often called the Father of Modern Medicine, wrote all parts of the body which have a function, if used in moderation and exercised in labours in which each is accustomed.

2.1. Physical Activity and Hypokinetic Diseases/ Conditions

Hypokinetic diseases are conditions related to inactivity or low levels of habitual activity.

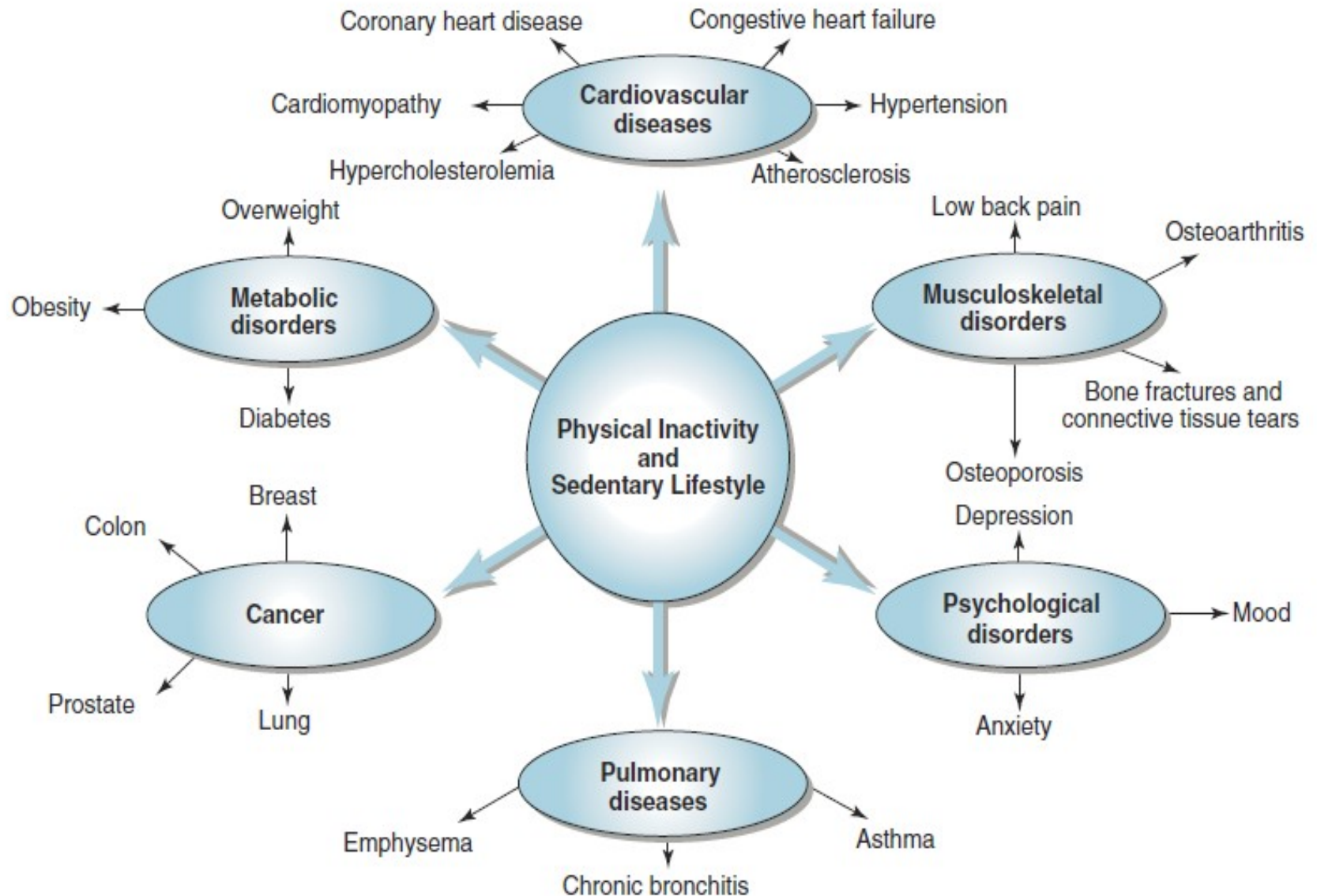
The term “hypokinetic” was coined by Kraus and Raab in their book *Hypokinetic Disease* (Kraus & Raab, 1961).

This term is now widely accepted and can be used to describe many of the diseases and conditions associated with inactivity and poor fitness such as those conditions outlined in *Physical Activity and Health: A Report of the Surgeon General* (USDHSS, 1996).

Individuals who do not exercise regularly are at a greater risk for developing chronic diseases such as:-

- coronary heart disease (CHD),
- hypertension,
- hypercholesterolemia,
- cancer,
- Metabolic disordered
- Psychological disordered
- obesity, and
- musculoskeletal disorders

Role of physical activity and exercise in disease prevention and rehabilitation



Exercise scientists and health and fitness professionals have maintained that regular physical activity is the best defence against the development of many diseases, disorders, and illnesses.

Scientific evidence has increased our understanding of the benefits of physical activity for improved health and quality of life.

The intensity of exercise is expressed in metabolic equivalents (METs).

A MET is the ratio of the person's working (exercising) metabolic rate to the resting metabolic rate.

One MET is defined as the energy cost of sitting quietly.

Summary of the ACSM and AHA physical activity recommendations for adults

	AEROBIC ACTIVITY ^a			MUSCLE-STRENGTHENING ACTIVITY			FLEXIBILITY/ BALANCE ACTIVITY
Population group	Duration ^b (min/day)	Intensity	Frequency (days/wk)	Sets	Intensity or # of exercises	Frequency (days/wk)	
Healthy adults 18-65 yr	30	Moderate (3.0–6.0 METS)	Minimum 5	1	8- to 12-RM; 8–10 exercises for major muscle groups	≥2 noncon- secutive days	No specific recommendation
	20	Vigorous (>6.0 METS)	Minimum 3				
Older adults >65 yr	30	Moderate (5 or 6 on 10 pt. scale)	Minimum 5	1	10- to 15-RM; 8–10 exercises for major muscle groups;	2 nonconsec- utive days	For flexibility at least 2 days per wk for at least 10 min each day; include balance exer- cises for those at risk for falls.
	20	Vigorous (7 or 8 on 10 pt. scale)	Minimum 3		Moderate intensity (5 or 6 on 10 pt. scale) Vigorous intensity (7 or 8 on 10 pt. scale)		

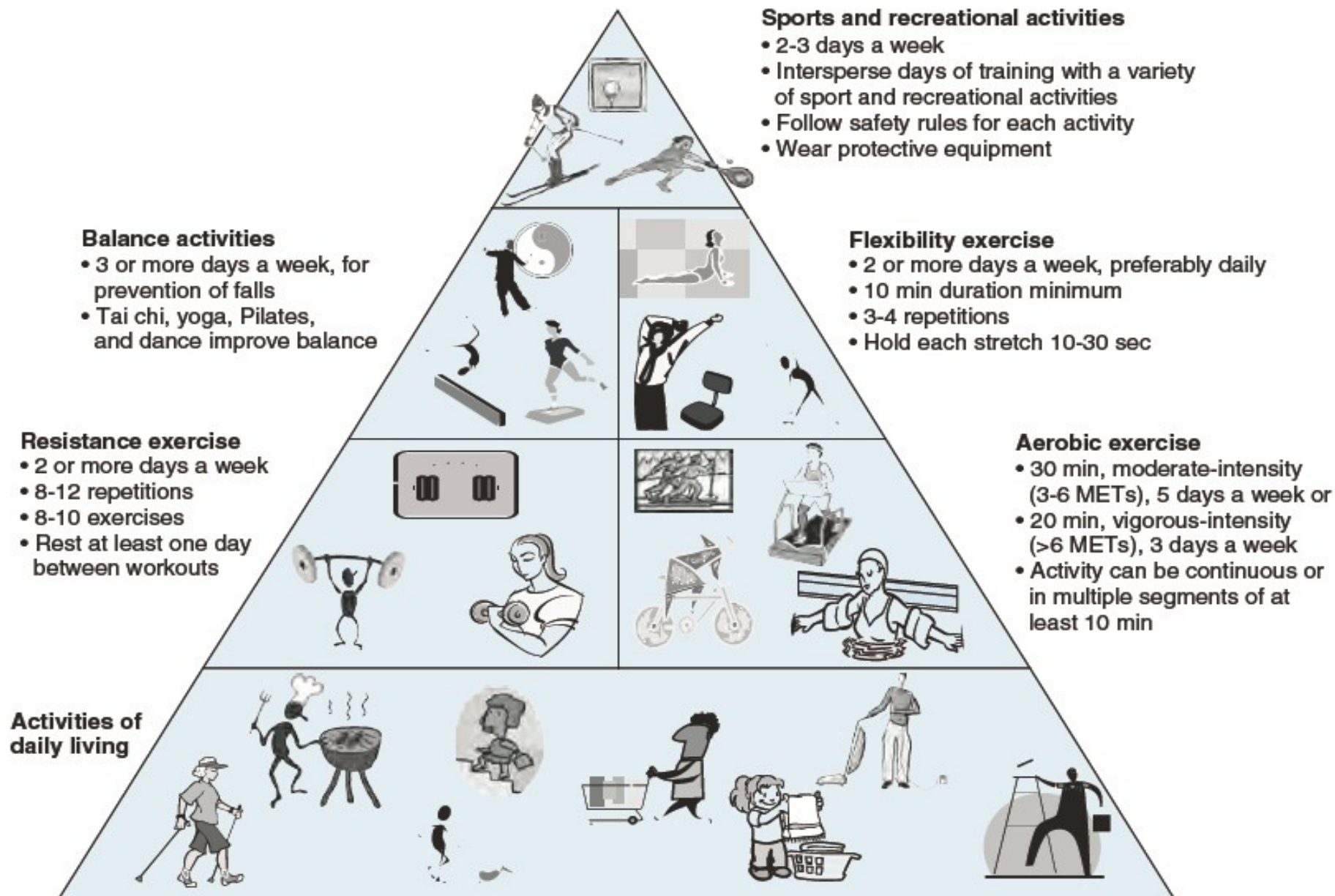
- Improvements in health benefits depend on the volume (i.e., combination of frequency, intensity, and duration) of physical activity. This is known as the dose-response relationship.
- Physical activity lowers the risk of hypokinetic conditions including:-
 - dying prematurely,
 - coronary artery disease,
 - stroke,
 - type 2 diabetes,
 - metabolic syndrome,
 - high blood lipid profile,
 - cancers (colon, breast, lung, and endometrial), and
 - hip fractures

- For adults (18–65 yr) and older adults (>65 yr), the ACSM recommends a minimum of 30 min of moderate-intensity aerobic activity 5 days per week or 20 min of vigorous-intensity aerobic exercise 3 days per week.
- They also recommend moderate- to high-intensity (8- to 12-repetition maximum (RM) for adults and 10- to 15-RM for older adults) resistance training for a minimum of 2 non-consecutive days per week.
- Balance and flexibility exercises are also suggested for older adults.

- The Exercise and Physical Activity Pyramid illustrates a balanced plan of physical activity and exercise to promote health and to improve physical fitness.
- You should encourage your clients to engage in physical activities around the home and workplace on a daily basis to establish a foundation (base of pyramid) for an active life style.
- They should perform aerobic activities a minimum of 3 days/wk; they should do weight-resistance exercises and flexibility or balance exercises at least 2 days per week.

- Low intensity physical activities (bottom of pyramid) require light physical exercise.
- Recreational sport activities (middle levels of pyramid) are recommended to add variety to the exercise plan.
- High-intensity training and competitive sport (top of pyramid) require a solid fitness base and proper preparation to prevent injury; most adults should engage in these activities sparingly

The Exercise and Physical Activity Pyramid



Physical Activity and Cardiovascular Diseases

- According to World Health Organization (WHO, 2011) cardiovascular disease (CVD) caused 17.3 million deaths (30%) worldwide in 2008, and it is projected to cause more than 26 million deaths by 2030.
- More than 80% of those cardiovascular deaths occurred in low- and middle-income countries.

Coronary Heart Disease (CHD)

- CHD is caused by a lack of blood supply to the heart muscle (myocardial ischemia) resulting from a progressive, degenerative disorder known as atherosclerosis.
- Atherosclerosis is an inflammatory process involving a build-up of low-density lipoprotein (LDL) cholesterol, scavenger cells (monocytes), necrotic debris, smooth muscle cells, and fibrous tissue.
- This is how plaques form in the intima, or inner lining of the medium- and large-sized arteries throughout the cardiovascular system

- Coronary Heart Disease Risk Factors: The greater the number and severity of risk factors, the greater the probability of CHD. The positive risk factors for CHD are:
 - Age,
 - Family History,
 - Hypercholesterolemia,
 - Hypertension,
 - Tobacco use,
 - Diabetes Mellitus or Prediabetes,
 - Overweight and Obesity, and
 - Physical Inactivity.

- Physical Activity and Coronary Heart

Physically active people have lower incidences of myocardial infarction and mortality from CHD and tend to develop CHD at a later age compared to their sedentary counterparts.

Individuals who exercise regularly reduce their relative risk of developing CHD by a factor of 1.5 to 2.4. Leading a physically active lifestyle may prevent 20% to 35% of cardiovascular diseases.

Hypertension

- Hypertension, or high blood pressure, is a chronic, persistent elevation of blood pressure that is clinically defined as a systolic pressure ≥ 140 mmHg or a diastolic pressure ≥ 90 mmHg.
- Regular physical activity prevents hypertension and lowers blood pressure in younger and older adults who are normotensive, prehypertensive, or hypertensive

Hyper-cholesterolemia and Dyslipidemia

- Hypercholesterolemia, is an elevation of total cholesterol (TC) in the blood, is associated with increased risk for CVD.
- Hypercholesterolemia is also referred to as hyperlipidemia, which is an increase in blood lipid levels;
- dyslipidemia refers to an abnormal blood lipid profile.
- Age, gender, family history, alcohol, smoking are risk factors for hypercholesterolemia and regular activity reduced the chance of getting hypercholesterolemia and dyslipidemia.

- LDLs, HDLs, and TC: Cholesterol is a waxy, fatlike substance found in all animal products (meats, dairy products, and eggs). The body can make cholesterol in the liver and absorb it from the diet.
- Lipoproteins are classified by the thickness of the protein shell that surrounds the cholesterol.

- The four main classes of lipoproteins are chylomicron, derived from the intestinal absorption of triglycerides (TG);
- very low-density lipoprotein (VLDL), made in the liver for the transport of triglycerides;
- low-density lipoprotein (LDL), a product of VLDL metabolism that serves as the primary transporter of cholesterol; and
- high-density lipoprotein (HDL), involved in the reverse transport of cholesterol to the liver.

Diabetes Mellitus

Diabetes is a global epidemic. More than 346 million people worldwide have the disease (WHO, 2011).

Factors linked to this epidemic include urbanization, aging, physical inactivity, unhealthy diet, and obesity.

At least 65% of people with diabetes mellitus die from some form of heart or blood vessel disease (AHA, 2008).

Diabetes is a major contributor toward the development of CHD, stroke and kidney failure .

- Pre-diabetes, in addition to being a positive risk factor for CVD, is a medical condition identified by fasting blood glucose or glycated, hemoglobin (HbA1c) levels that are above normal values but lower than the threshold for a diagnosis of diabetes.
- Fortunately, pre-diabetes appears to respond favourably to weight loss and increases in physical activity.

- **Type 1 diabetes**, formerly referred to as insulin-dependent diabetes mellitus (IDDM), usually occurs before age 30 but can develop at any age.
- **Type 2 diabetes**, previously known as non-insulin-dependent diabetes mellitus (NIDDM), is more common; 90% of individuals diagnosed with diabetes mellitus worldwide have type 2 diabetes (WHO, 2011). Age, gender, family history, calorie intake, physical inactivity are risk factors for developing diabetes.

Type 1 diabetes may be caused by autoimmune, genetic, or environmental factors, but the specific cause is unknown. Unfortunately, there is no known way to prevent type 1 diabetes (CDC, 2011).

Healthy nutrition and increased physical activity, however, can reduce the risk of type 2 diabetes by as much as 67% in high-risk individuals.

Nearly 90% of cases of type 2 diabetes worldwide may be related to obesity

- The effect of exercise alone as an intervention for people with type 2 diabetes is not well researched. However, exercise (30–120 min, 3 days/wk for 8 wk) produced clinically significant improvements in HbA1c and reduced visceral and subcutaneous adipose tissue stores in people with type 2 diabetes.
- Both resistance and aerobic exercise alone or in combination improve HbA1c values in people with type 2 diabetes

Obesity and Overweight

- overweight and obesity are classified using the body mass index(BMI) ($BMI = \text{weight [kg]} / \text{height squared [m}^2\text{]}$).
- Individuals with a BMI between 25 and 29.9 kg/m² are classified as overweight; those with a BMI of 30 kg/m² or more are classified as obese.
- Age, gender, family history, cholesterol intake, and physical inactivity are the major factors associated with increased risk of obesity.

- Excess body weight and fatness pose a threat to both the quality and duration of one's life.
- Obese individuals have a shorter life expectancy and greater risks of CHD, stroke, dyslipidemia, hypertension, diabetes mellitus, certain cancers, osteoarthritis, sleep apnea, abnormal menses, and infertility.
- Restricting caloric intake and increasing caloric expenditure through physical activity and exercise are effective ways of reducing body weight and fatness while normalizing blood pressure and blood lipid profiles

Metabolic Syndrome

- Metabolic syndrome refers to a combination of CVD risk factors associated with hypertension, dyslipidemia, insulin resistance, and abdominal obesity.
- Age and BMI directly relate to metabolic syndrome
- Lifestyle must be modified in order to manage metabolic syndrome.
- The combination of healthy nutrition and increased physical activity is an effective way to increase HDL-C and to reduce blood pressure, body weight, triglycerides, and blood glucose levels.

- A sedentary lifestyle and lack of physical activity reduce life expectancy by predisposing the individual to aging-related diseases and by influencing the aging process itself.
- With aging, a progressive loss of physiological and metabolic functions occurs; however, biological aging may differ considerably among individuals due to variability in genetic and environmental factors that affect oxidative stress and inflammation.
- regular exercise benefits in retarding the aging process and diminishing the risk of aging-related diseases.

Physical Activity and Postural Deformity

- Posture is the position from which movement begins and ends.
- Having proper postural alignment enables the body to perform movements quicker with less joint and muscular strain.
- The body is designed to work at the most economical level, thus saving energy for future use.
- We spend more energy maintaining misaligned posture, which can cause muscle and joint pain.

- Women in general tend to develop poor posture because of many factors.
- They often have more clerical and computer oriented jobs that require sitting in a chair, eyeing a computer screen for long periods of time.
- They also wear high-heeled shoes, which lead to an alteration and compensation of their posture.
- To improve your posture and reduce structural damage, you should adhere to a corrective postural exercise program

Exercises for correcting posture:

- Prone Cobra
- Axial Extension Trainer
- Wall Leans
- Cervical Extension using a blood pressure cuff

Musculoskeletal Diseases and Disorders

- Diseases and disorders of the musculoskeletal system, such as osteoporosis, osteoarthritis, bone fractures, connective tissue tears, and low back syndrome, are also related to physical inactivity and a sedentary lifestyle.
- Osteoporosis is a disease characterized by the loss of bone mineral content and bone mineral density due to factors such as aging, amenorrhea, malnutrition, menopause, and physical inactivity

- Osteopenia, or low bone mineral mass, is a precursor to osteoporosis.
- Adequate calcium intake, vitamin D intake, and regular physical activity help counteract age-related bone loss.
- Exercise Prescription for Preserving Bone Health of Adults
 - Mode
 - Intensity
 - Frequency
 - Duration

Low back pain

- Low back pain afflicts millions of people each year. More than 80% of all low back problems are produced by muscular weakness or imbalance caused by a lack of physical activity.
- the muscles are not strong enough to support the vertebral column in proper alignment, poor posture results and low back pain develops.
- Excessive weight, poor flexibility, and improper lifting habits also contribute to low back problems

- While some risks of associated with low back pain are not modifiable, such as gender and age, lifestyle behaviour such as smoking, physical inactivity, flexibility, and muscular strength and endurance can all be improved.
- Because the origin of low back problems is often functional rather than structural, in many cases, the problem can be corrected through an exercise program that develops strength and flexibility in the appropriate muscle groups